



CONTROLLED DRAFT

System Architecture & Technology Specification

Production architecture, module boundaries, stack and dependency rules

Product	Nod
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Owner / operator	[NOD OWNER / LEGAL ENTITY]
Status	Controlled Draft

Release note: This pack is a production-ready baseline. Bracketed operator details must be completed before public release. Legal documents should be reviewed by Nigerian counsel before publication or execution.

1. Technology baseline

Layer	Selected technology
Language	Kotlin
UI	Jetpack Compose + Material 3 foundation
State	ViewModel + StateFlow / unidirectional data flow
Concurrency	Kotlin Coroutines + Flow
DI	Hilt + KSP
Database	Room
Preferences	Proto DataStore
Control protocol	Protocol Buffers
Primary transport	Google Nearby Connections - P2P_POINT_TO_POINT
Future transport	Wi-Fi Direct adapter behind NodTransport
Security	Android Keystore + AES-GCM; SHA-256 integrity
Camera/OCR	CameraX + bundled ML Kit Text Recognition + libphonenumber
Background	Foreground Service + notification
Quick launch	TileService + accelerometer SensorManager
Storage	MediaStore + Storage Access Framework + private temp storage
Testing	JUnit, coroutine test, Compose tests, Robolectric, macrobenchmark and physical-device matrix

2. Layered architecture

Compose UI -> ViewModels -> domain use cases -> repositories -> platform/transport adapters -> Android OS and local storage. UI modules must not call Nearby Connections, Room or Android providers directly.

3. Dependency rules

- feature modules depend on domain abstractions and design system.
- domain modules contain no Android UI or SDK-specific transport types.
- transport:api defines NodTransport and transport-neutral events.
- transport:nearby implements transport:api using ConnectionsClient.
- data modules implement repositories and are injected through interfaces.
- platform modules wrap contacts, files, sensors, permissions, notifications and Quick Settings APIs.

4. Suggested project structure

app/; core/common, design-system, model, protocol, security, testing; data/profile, transfer, history, settings; domain/discovery, session, transfer, profile, scanner; transport/api, nearby, wifi-direct; platform/contacts, files, permissions, sensors, notifications, quick-settings; feature/onboarding, home, nearby, connection, send, receive, scanner, history, profile, settings.

5. Primary domain interfaces

- NodTransport: advertise, discover, connect, send control bytes, send file payloads, disconnect, observe events.
- SessionManager: peer lifecycle, verification, trust state, protocol negotiation, timeout and disconnect.
- TransferManager: manifest, acceptance, capacity checks, payload mapping, progress, hashing, cancellation and finalization.
- ProfileRepository: encrypted Nod Profile and Share Profiles.
- HistoryRepository: minimal completed/failed transfer metadata.
- QuickNodManager: orchestrates shake/tile/manual launch into the same temporary discovery session.

6. Threading and concurrency

- All transport callbacks are converted to structured Flows at adapter boundaries.
- Disk I/O and hashing run on bounded background dispatchers.
- UI collects immutable state; it never owns transfer sockets/clients.
- Cancellation propagates from domain jobs to payload operations and cleanup.
- Backpressure must prevent progress/event floods from overwhelming the main thread.

7. Build configuration

- minSdk: 26.
- targetSdk: 36 for new Google Play release after 31 August 2026.
- compileSdk: current stable compatible with target 36 at release.
- JDK 17 toolchain.
- Release builds shrink/optimize with R8 and keep rules tested for Protobuf/ML Kit dependencies.
- Release signing secrets never committed to source control.

8. Architecture decisions

ADR	Decision	Reason
ADR-001	Native Android/Kotlin	Nod is deeply coupled to Android wireless, sensors, background, storage and permission models.
ADR-002	Local-first identity	Avoid mandatory account/backend and minimize privacy surface.
ADR-003	Transport abstraction	Protect product architecture from GMS-only lock-in.
ADR-004	Protobuf control plane	Strong schema/versioning for device interoperability.
ADR-005	Separate control/data plane	Small commands remain independent from large file payload mechanics.
ADR-006	Contextual permissions	Request only when feature requires capability.

Authoritative references

These links are included for implementation and legal review. Requirements can change; verify them again immediately before release.

- **Nigeria Data Protection Act 2023:** <https://ndpc.gov.ng/download/nigeria-data-protection-act-2023>
- **NDP Act General Application and Implementation Directive (GAID) 2025:** <https://ndpc.gov.ng/wp-content/uploads/2025/07/NDP-ACT-GAID-2025-MARCH-20TH.pdf>
- **Nigeria Data Protection Commission resources:** <https://ndpc.gov.ng/resources/>
- **Copyright Act 2022 - Nigerian Copyright Commission:** <https://www.copyright.gov.ng/CopyrightAct/>
- **Nigerian Copyright eRegistration System:** <https://eregistration.copyright.gov.ng/>
- **Google Play - User Data policy:**
<https://support.google.com/googleplay/android-developer/answer/10144311>
- **Google Play - Prepare your app for review:**
<https://support.google.com/googleplay/android-developer/answer/9859455>
- **Google Play - Target API requirements:**
<https://support.google.com/googleplay/android-developer/answer/11926878>
- **Google Play - Foreground service requirements:** <https://support.google.com/googleplay/android-developer/answer/13392821>